Closure properties of Context Free language

- context Free languages are closed under:
  (a) Union (b) Concatenation (c) Kleene closure

Not closed under:
  (a) Intersection (b) Complementation

Note: closed means if L_1 & L_2 is CFL, then result of both also gives CFL

(1) Union ($\cup$)

- combine two languages

example: $L_1 = \{a^n b^n\} \rightarrow ab, aabb, aaabbb, \ldots$

$L_2 = \{a^n c^n\} \rightarrow ac, aacc, aaaccc, \ldots$

Union: $L_1 \cup L_2 = \{ab, aabb, aaabbb, ac, aacc, aaaccc \ldots\}$

$\rightarrow$ means strings can belong to either L_1 or L_2

(2) Concatenation ($\cdot$)

$L_1 = \{a^n b^n\} \rightarrow ab, aabb, \ldots$

$L_2 = \{c^n\} \rightarrow c, cc, ccc, \ldots$

Concatenation: $L_1 \cdot L_2$

$= \{abc, abcc, abccc, aabbc, aabbcc, \ldots\}$

(3) Kleene Star / Kleene closure

if L is a CFL, L^* is also a CFL

ex $L = \{a\}$

$L^* = \{\varepsilon, a, aa, aaa, \ldots\}$

(4) Intersection

if L_1 and L_2 are two context free languages, their intersection $L_1 \cap L_2$ need not be context free.

ex. $L_1 = \{a^n b^n c^m \mid n \geq 0, m \geq 0\}$

$L_2 = \{a^m b^n c^n \mid n \geq 0, m \geq 0\}$

$L_3 = L_1 \cap L_2 = \{a^n b^n c^n \mid n \geq 0\}$

L_3 need not be context free

(5) Complement / Complementation

if L_1 is context free language, complement of L_1 is $= \Sigma^* - L_1$, which need not be context free.